

ASHRAF SIDDIQUE

Computer Vision & AI Researcher

Experience

- 2025–present: **Postdoctoral Researcher, UNIST, Ulsan, South Korea.**
Feed Forward Multi-View 3D Reconstruction, Machine Learning and Deep Learning
- 2022–2025: **AI and Computer Vision Engineer, OPIN INC., Suwon, South Korea.**
- Led development of object detection, segmentation, and anomaly detection systems for industrial automation
 - Built multi-view 3D reconstruction pipelines integrating calibration, sensor fusion, and pose estimation
 - Designed 2D/3D alignment algorithms for precision robotics and quality inspection
- 2013–2015: **Technical Analyst, Standard Chartered Bank, Bangladesh.**
- Upgraded and maintained core banking systems
 - Resolved application issues across financial platforms
 - Supported in-house applications to ensure high system reliability

Education

- 2015–2022: **MS and PhD, Computer Science & Engineering, Kyung Hee University, South Korea.**
Thesis Title: Object-wise Video Synthesis and Editing in Epipolar Geometry of 3D Space.
Research areas: 3D reconstruction, semantic segmentation, inpainting, object removal, super-resolution, deep learning, geometry-based vision
- 2009–2013: **BIT, Information Technology, IIT, University of Dhaka, Bangladesh.**
Core foundation in algorithms, data structures, networking, and database systems

Publications

Journal Articles

- 2024 Sumiaya Rahman, Hyun-Jin Jeong, **Siddique, Ashraf**, Yong-Jae Moon, and Bendict Lawrance. Near-real-time 3d reconstruction of the solar coronal parameters based on the magnetohydrodynamic algorithm outside a sphere using deep learning. *The Astrophysical Journal Supplement Series*, volume 271, page 14. IOP Publishing, 2024, (**Impact Factor:9.2**).
- 2023 Sumiaya Rahman, **Siddique, Ashraf**, Seungheon Shin, Hyun-jin Jeong, Yong-Jae Moon, Eunsu Park, Jihye Kang, and Sung-Ho Bae. Fast reconstruction of 3d density distribution around the sun based on the mas by deep learning. *The Astrophysical Journal*, volume 948, page 21. IOP Publishing, 2023, (**Impact Factor:4.8**).
- 2022 **Siddique, Ashraf** and Seungkyu Lee. Sym3dnet: Symmetric 3d prior network for single-view 3d reconstruction. *Sensors*, volume 22, page 518. MDPI, 2022, (**Impact Factor:3.7**).
- 2021 **Siddique, Ashraf** and Seungkyu Lee. Object-wise video editing. *Applied Sciences*, volume 11, page 671. MDPI, 2021, (**Impact Factor:2.7**).

- 2020 Sumiaya Rahman, Yong-Jae Moon, Eunsu Park, **Siddique, Ashraf**, Il-Hyun Cho, and Daye Lim. Super-resolution of sdo/hmi magnetograms using novel deep learning methods. *The Astrophysical Journal Letters*, volume 897, page L32. IOP Publishing, 2020, (**Impact Factor:11.7**).
- 2020 Md Imtiaz Hossain, **Siddique, Ashraf**, Md Alamgir Hossain, Md Delowar Hossain, and Eui-Nam Huh. Batch entropy supervised convolutional neural networks for feature extraction and harmonizing for action recognition. *IEEE Access*, volume 8, pages 206427–206444. IEEE, 2020, (**Impact Factor:3.6**).
- In Conference Proceedings
- 2024 Sumiaya Rahman, Hyun-Jin Jeong, **Siddique, Ashraf**, and Yong-Jae Moon. Near real-time construction of solar coronal parameters based on mas simulation by deep learning. In *EGU General Assembly Conference Abstracts*, page 14186, 2024.
- 2021 Sumiaya Rahman, Seungheon SHIN, Hyun-Jin Jeong, Yong-Jae Moon, and **Siddique, Ashraf**. Fast construction of 3-d solar coronal density distribution based on mas simulation by deep learning. In *AGU Fall Meeting Abstracts*, volume 2021, pages SH25F–2156, 2021.
- 2018 **Siddique, Ashraf** and Seungkyu Lee. Video inpainting for arbitrary foreground object removal. In *2018 IEEE Winter Conference on Applications of Computer Vision (WACV)*, pages 1755–1763. IEEE, 2018.
- 2016 **Siddique, Ashraf** and Seungkyu Lee. Video inpainting for object removal. pages 1078–1080, 2016.

Research Experience

OPIN INC

2024 – 2025 **PILLSTRIP-INSPECTOR.**

PillStrip-Inspector automates the verification of medicine strips to ensure each meal's pills are correct. It uses object detection and segmentation to detect pills and compares them to a reference image for accuracy

([Web-page](#))

2023 – 2024 **3D Alignment for Precision Logo Printing on Baseballs.**

Baseball Alignment ensures precise logo placement on baseballs using a multi-camera setup and 3D alignment framework. It captures the ball's orientation, calculates rotation angles, and adjusts the ball for accurate logo printing.

([Web-page](#)) ([Demo](#))

Sun & Space Weather Lab, Kyung Hee University

2019 – 2021 **Machine Learning for Solar Coronagraph and Space Weather Research.**

Contributed machine learning modules to two major projects under the Development of a Solar Coronagraph on the ISS program. Developed deep learning models for solar coronal imaging and feature extraction, resulting in publications in **The Astrophysical Journal Letters** and **The Astrophysical Journal Supplement Series**.

Supervisor : **Prof. Yong-Jae Moon**, Professor, Department of Astronomy and Space Science, Kyung Hee University, South Korea

Perception & Computer Vision Lab, Kyung Hee University

2019 – 2021 **Single-View 3D object Reconstruction.**

We propose Sym3DNet for single-view 3D object reconstruction, which employs a 3D reflection symmetry structure prior of an object. On the ShapeNet13 dataset, it achieves a 0.68 IoU and 0.44 F-score.

Advisor : **Prof Seungkyu Lee**, Professor, Department of Computer Science & Engineering, Kyung Hee University

2016 – 2020 **OBJECT-WISE VIDEO EDITING.**

We introduce a 3D-based video editing method that calculates scene details like camera parameters and depth using techniques like SFM, SLAM, and bundle adjustment. Our approach removes objects, inpaints the video, stabilizes it, and creates new views. Our inpainting work was published in WACV 2018. ([Link](#))

Advisor : **Prof Seungkyu Lee**, *Professor, Department of Computer Science & Engineering*, Kyung Hee University

Computer skills

Programming Python, C++, CUDA, PyTorch, TensorFlow

3D Vision VGGT, Depth Anything, Neural Radiance Fields (NeRF), Multi-view Geometry, SLAM

Computer Vision Semantic Segmentation, Object Detection, Anomaly Detection, OCR, Camera Calibration, Feature Matching (LoFTR, LightGlue), Point Cloud Processing

Frameworks PyTorch, TensorFlow, Hugging Face Transformers

Graphics Blender, Open3D, COLMAP, Mesh Processing

Tools OpenCV, GitHub, Docker